

Vishal Kachhap

Full Stack Developer — Software Engineer

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PROFESSIONAL SUMMARY

Full Stack Developer with 5+ years of experience building production-grade web applications, REST APIs, data-driven platforms, and cloud-deployed systems. Experienced in Python, TypeScript, Angular, React, Node.js, GCP, Docker, distributed systems, and database optimization, with experience delivering engineering and financial applications.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL **Frontend:** Angular, React.js, Next.js, Plotly
Backend: Node.js, Express.js, NestJS, Flask, REST APIs **Cloud/DevOps:** GCP, Docker, Microservices
Databases: MongoDB, Database Optimization **Messaging:** Apache Kafka, Amazon SQS
Architecture: Scalable System Design, Distributed Systems **Security/Tools:** Authentication & Authorization, Git, GitHub Copilot

PROFESSIONAL EXPERIENCE

Oorja 2024 – Present

Project: Battery Engineering Simulation Platform

- Developed and integrated physics-based mathematical models and simulations into a production application used by 20–30 R&D engineers for compute-intensive battery engineering workflows.
- Developed up to 80% of the application's REST APIs using Python, enabling data retrieval, processing, analysis, and integration between engineering models and application services.
- Optimized database queries and data-processing workflows across millions of records, reducing API/query latency by approximately 40% and processing time by 40–50%.
- Led Angular frontend development and implemented interactive Plotly visualizations for simulation results, engineering plots, and KPIs, supporting up to 10 simulation models per user.
- Developed a model-copy feature that allows users to duplicate existing models with similar configurations, reducing model setup time from 2–3 hours to a few minutes and significantly simplifying repetitive engineering workflows.
- Managed application infrastructure and deployment using Docker and GitHub workflows, automating Docker image builds, publishing images to Google Artifact Registry (GAR), and deploying containerized applications on Cloud Run.

Tech Stack: Python, Angular, Plotly, REST APIs, GCP, Docker, Google Artifact Registry, Cloud Run, SQL/MongoDB, Git
Better Wealth Technologies Pvt. Ltd. 2021 – 2023

Project: Real-Time Financial Alerting System

- Designed and deployed a real-time financial alerting system using Amazon SQS and Twilio to asynchronously process and deliver up to 10,000 notifications per day to users.
- Built reliable event-driven notification Actions to deliver financial alerts to the platform's user base.

Tech Stack: Node.js, Amazon SQS, Twilio, REST APIs

Project: Stock Advisory Platform

- Led development of an interactive stock advisory platform used by 20 experts and reaching 10,000 followers, with data coverage across approximately 10,000 stocks.
- Built advanced data tables with filtering, real-time search, pagination, and sorting to enable efficient exploration of large stock datasets.
- Developed user-facing forms and integrated Google Analytics and Google Tag Manager with custom events and conversion tracking to measure user engagement and platform interactions.

Tech Stack: React.js, Node.js, JavaScript, Google Analytics, Google Tag Manager

Project: Admin Dashboard

- Developed and maintained a React.js and Node.js admin dashboard with authentication, data visualization, and CRUD functionality for user and data management.

Tech Stack: React.js, Node.js, MongoDB, REST APIs, Authentication, CRUD

EDUCATION

IIT Kharagpur – B.Tech. in Mechanical Engineering

2017 – 2021